

AXISFIELD-GS: FEED-FORWARD ARTICULATION RECOVERY FROM SINGLE-STATE SEMANTIC GAUSSIAN ASSETS

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ABSTRACT

Static 3D Gaussian assets capture object appearance but lack the kinematic structure needed for interaction. We present AxisField-GS, a feed-forward method that recovers a primary revolute joint and its movable region from a single-state semantic Gaussian asset. The method operates directly on an existing asset, without additional motion observations or per-object optimization. Each Gaussian predicts an axis offset and an unoriented direction, together with movable confidence. A self-consistent voting rule combines these cues to select reliable anchors for a differentiable line fit, linking local predictions to the recovered joint. On 632 PartNet-Mobility-derived test assets from unseen objects, it achieves 0.063 normalized symmetric axis distance, 2.43° angular error, and 0.864 movable IoU. Same-checkpoint comparisons and matched retraining support the benefits of geometric vote selection, while rotation tests expose the dependence on canonical object orientation.

Index Terms— 3D Gaussian splatting, articulation recovery, dense voting, geometric estimation

1. INTRODUCTION

3D Gaussian Splatting (3DGS) represents photorealistic digital objects [1]. Semantic extensions augment Gaussians with distilled visual or language features for point-level understanding, segmentation, selection, and editing [2, 3, 4]. Yet static assets do not specify how their parts move. We study recovering a primary revolute axis and movable Gaussian region from an existing single-state semantic asset.

Prior work infers articulation from shapes [5], interactions [6], or multiple configurations [7]. Category-level approaches estimate articulated pose and joint parameters from single-depth observations [8]. GPartNet studies actionable parts across categories [9]. Particulate predicts articulation from a static mesh [10]; S2O enhances static objects with openable structure [11].

Gaussian methods ArtGS [12], ArticulatedGS [13], and SplArt [14] couple articulation with multi-state reconstruction. SPAGS reconstructs and optimizes an object from sparse

views of one state [15]. ArtSplat performs feed-forward articulated Gaussian reconstruction from sparse multi-state observations [16]. Our setting starts with an existing single-state semantic Gaussian asset and learns an articulation predictor across objects, without additional motion observations or per-object optimization. The predicted movable field remains attached to the original Gaussians.

Without observed motion, a broad movable surface may identify the part while providing weak evidence of hinge location. A revolute axis is also unchanged by shifting its origin along the line or reversing its direction. We therefore aggregate local geometric evidence on a common unoriented line.

AxisField-GS predicts dense anchors and local directions, then fits a line weighted by movable confidence, offset proximity, and directional agreement. Building on established dense voting [17, 18], we formulate primary articulation recovery on semantic Gaussian assets as aggregating local geometric evidence into a globally consistent revolute line. Experiments separate readout diagnostics from retrained ablations and evaluate canonical-orientation dependence.

2. METHOD

2.1. Dense articulation fields

An input asset contains Gaussians with centers $p_i \in \mathbb{R}^3$, rotations, anisotropic scales, opacities, estimated normals, and DINOv2 features [19] compressed to 32 dimensions by PCA. Coordinates are normalized in the object frame while retaining the dataset’s canonical orientation. The output is a revolute line $\mathcal{A} = (c, v)$, with unit direction v , and a movable probability for each Gaussian. We consider one primary revolute joint per asset.

A Point Transformer V3 backbone [20] encodes a shortlist of 8192 Gaussians into multiscale memories. A three-layer decoder with four learnable joint queries aggregates object-level context. A point decoder then reads local memory neighborhoods and joint context to predict, for each Gaussian, movable probability m_i , auxiliary support probability b_i , an axis offset Δ_i , and a unit local direction d_i . The predicted

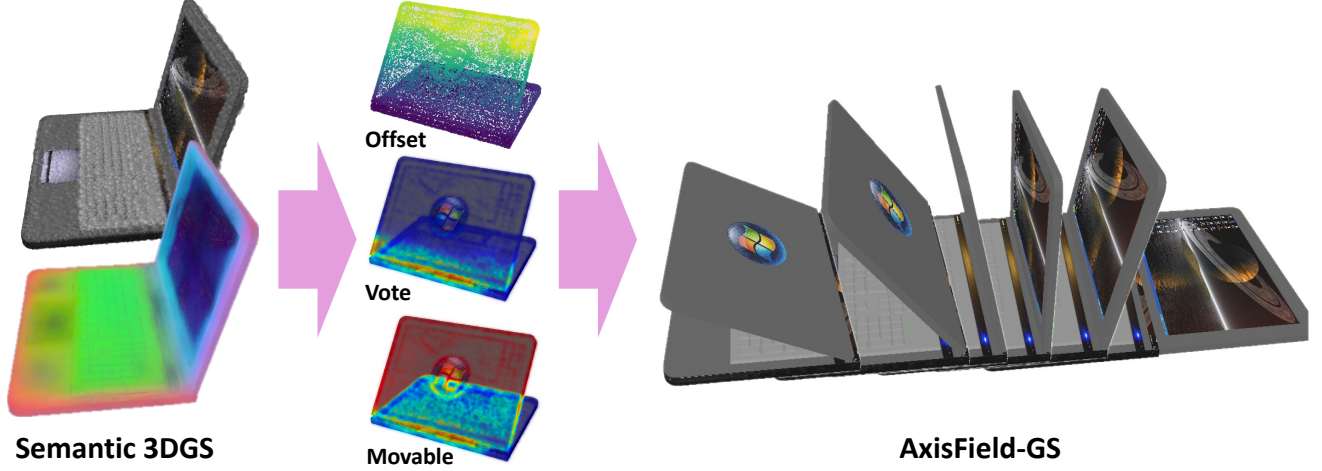


Fig. 1. From a static asset to articulation. AxisField-GS predicts dense fields from a single-state semantic Gaussian asset. Axis offsets, movable confidence, and local direction agreement determine the vote weights used to recover a revolute line. The recovered axis and movable field are applied to the original Gaussians to synthesize articulated states.

anchor is

$$a_i = p_i + \Delta_i. \quad (1)$$

Offsets are supervised toward the closest point on the ground-truth axis. The support target is a Gaussian band over movable Gaussians near that axis; it supplies auxiliary supervision and is not an inference-time vote mask. The backbone and attention modules provide standard point-set reasoning. The geometric estimator below determines how their dense outputs become an articulation axis.

2.2. Self-consistent line estimation

Movable confidence alone can give excessive influence to broad surfaces far from a hinge. We therefore first combine movable probability with offset proximity:

$$\tilde{w}_i = m_i \exp\left(-\frac{\|\Delta_i\|_2^2}{2\sigma_o^2}\right). \quad (2)$$

This term favors evidence originating near the predicted axis. To measure agreement without assigning an arbitrary direction sign, we form an orientation matrix and its leading eigenvector:

$$u = \text{eig}_{\max}\left(\sum_i \tilde{w}_i d_i d_i^\top\right), \quad w_i = \tilde{w}_i |d_i^\top u|. \quad (3)$$

Both $d_i d_i^\top$ and $|d_i^\top u|$ are invariant to direction reversal. Thus, high-weight votes are movable, close to the axis, and consistent with a shared orientation. The support prediction is not required by this rule.

Let S be the selected anchor set and $W = \sum_{i \in S} w_i + \epsilon$. The fitted line is given by

$$c = W^{-1} \sum_{i \in S} w_i a_i, \\ C = W^{-1} \sum_{i \in S} w_i (a_i - c)(a_i - c)^\top, \quad v = \text{eig}_{\max}(C). \quad (4)$$

The centroid supplies a representative point on the line, while the principal direction minimizes weighted orthogonal residuals. Local direction predictions select reliable anchors; the final axis direction is estimated from their spatial distribution. This separates evidence selection from line fitting and avoids regressing a unique hinge origin where none exists. Figure 1 illustrates the recovered articulation.

We use $\sigma_o = 0.08$. At inference, S contains the top 256 votes together with any additional votes of weight at least 0.05. The top-ranked set also supplies a fallback when no vote reaches the cutoff. The same predicted weighting and differentiable line fit are used during training, with dense query sampling to control memory use.

2.3. Learning the fields and the axis

Dense supervision combines focal binary cross-entropy for movable and support probabilities, a nearest-axis offset loss, and the sign-invariant direction loss $1 - |d_i^\top v^*|$. Their respective weights are 1, 1, 2, 1. The fitted axis receives an additional loss

$$\mathcal{L}_{\text{axis}} = 3d_{\text{sym}}((c, v), (o^*, v^*)) + 1.5(1 - |v^\top v^*|), \quad (5)$$

where (o^*, v^*) is the target axis and d_{sym} is defined in Sec. 3.1. This term is enabled after 60 warmup epochs so

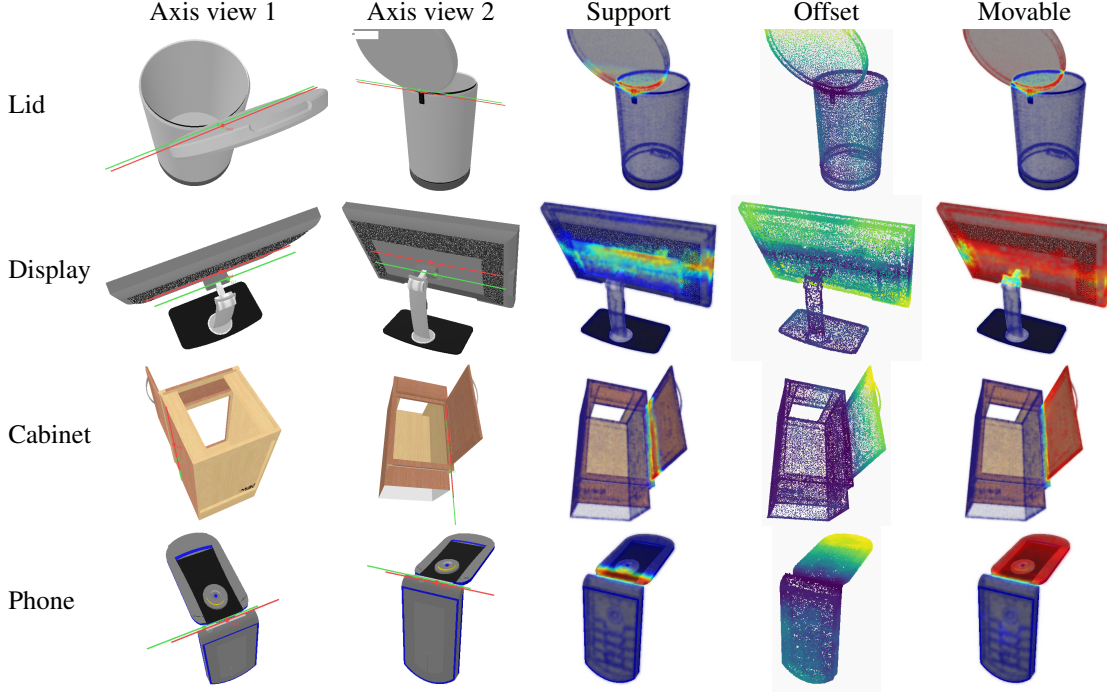


Fig. 2. Representative object-level visualizations of a lid, display screen, cabinet door, and flip phone (rows). Columns show two views of predicted (red) and ground-truth (green) revolute axes, followed by support, offset, and movable fields. Offset and movable predictions contribute to voting; support is an auxiliary prediction, not the vote weight.

that the dense predictions stabilize before supervising the fitted line. Auxiliary joint-query predictions provide joint-level supervision. Together, the objectives encourage locally accurate votes that also agree after aggregation.

The recovered line defines a rigid rotation for the movable Gaussians. Applying it to their centers and covariances produces the articulated states in Fig. 1, while leaving the static region fixed. This directly connects the estimated joint to the original asset without meshing or a separate rig.

3. EXPERIMENTS

3.1. Data and evaluation

We use single-state semantic Gaussian assets derived from PartNet-Mobility [21], covering nine categories: StorageFurniture, Laptop, TrashCan, Dishwasher, Safe, Display, WashingMachine, Microwave, and Phone. The training, validation, and test sets contain 1992, 256, and 632 assets. Validation and test contain 32 and 79 base objects, respectively. All states of a base object belong to the same split, so test objects are unseen during training. Each sample specifies one primary revolute target and its movable region; the model receives only the static asset.

For a unit direction v , define the point-to-line distance as $d_{\text{line}}(p; c, v) = \|(I - vv^\top)(p - c)\|_2$. We report

$$d_{\text{sym}} = \frac{1}{2} [d_{\text{line}}(c; o^*, v^*) + d_{\text{line}}(o^*; c, v)] \quad (6)$$

in normalized object coordinates, sign-invariant angular error $\arccos(|v^\top v^*|)$ in degrees, and movable IoU over valid Gaussians at probability threshold 0.5. All table entries are means unless specified otherwise.

We train with AdamW, learning rate 3×10^{-4} , weight decay 10^{-3} , batch size 64, and bfloat16 precision. The main model is trained for 1000 epochs; checkpoints are selected by validation symmetric distance. The point decoder uses 2048 sampled queries during training and processes the full asset in chunks of 4096 at inference. Separate 450-epoch runs provide matched retraining ablations, each with validation-based checkpoint selection.

3.2. Axis recovery and estimator comparison

Existing articulation methods use different input representations or require reconstruction, multi-state observations, or per-object optimization, so we focus on controlled comparisons under the fixed single-state semantic-Gaussian input setting. Table 1 summarizes overall performance.

Table 2 compares readouts of the same checkpoint, all retaining the movable head. The global estimate uses the highest-existence joint query; unweighted PCA fits predicted anchors without confidence weighting. Voting reduces symmetric distance by 55.8% over the jointly trained global head. The poor unweighted fit further shows the importance of vote selection for this checkpoint. These diagnostics isolate the

Table 1. Overall performance of the 1000-epoch main model. Dashes denote statistics not reported.

Split / statistic	Sym. ↓	Angle ↓	IoU ↑
Validation mean	0.0667	2.87°	0.894
Test mean	0.0631	2.43°	0.864
Test median	0.0331	—	—
Test success: 86.4% at Sym. < 0.10 and 96.7% at angle < 10°.			

Table 2. Same-input, same-checkpoint axis readouts. The auxiliary global head is jointly trained with the dense fields.

Estimator	Sym. ↓	Angle ↓	IoU ↑
Auxiliary global head	0.143	2.76°	0.864
Unweighted anchor PCA	0.413	39.94°	0.864
Self-consistent voting	0.063	2.43°	0.864

readout; they do not compare independently trained architectures.

Figure 2 contrasts broad movable fields with concentrated near-axis support. Broad doors remain harder: their mean distance is 0.108, versus 0.037 for cabinet doors and 0.056 for lids. Ambiguous hinge sides and weak local structure can shift an otherwise plausible axis.

3.3. Semantic features and vote selection

Removing semantic features during training and inference worsens symmetric distance from 0.063 to 0.094, angular error from 2.43° to 4.18°, and movable IoU from 0.864 to 0.770.

At a fixed checkpoint, removing the movable, offset, or direction factor increases symmetric distance to 0.381, 0.134, or 0.180, respectively. To distinguish checkpoint reliance from indispensability, Table 3 retrains each variant with its factor omitted in both training and inference.

After retraining, removing offset proximity worsens both axis metrics and IoU. Axis intervals for movable and direction removal include zero, suggesting partial redundancy; direction removal nevertheless reduces IoU. Bootstrap intervals reflect test-object sampling, not training-seed variation. This separate 450-epoch control is compared only with variants in the same table.

3.4. Orientation dependence and limitations

A global modal direction from training axes yields 41.08° mean error; a category-conditioned mode with oracle labels yields 8.04°, versus 2.43° for AxisField-GS. Thus, category priors explain part of the angular accuracy, but do not localize the axis line.

Random rotations substantially degrade performance; SO(3) augmentation narrows the gap (Table 4) without pro-

Table 3. Matched 450-epoch retraining. †: the paired 95% object-bootstrap interval for the difference from Full excludes zero (1000 resamples).

Training variant	Sym. ↓	Angle ↓	IoU ↑
Full retraining	0.059	1.14°	0.871
w/o movable factor	0.061	1.39°	0.865
w/o offset factor	0.072†	2.41°†	0.832†
w/o direction factor	0.060	1.66°	0.853†

Table 4. Orientation robustness. Rotated results average three Haar-random test-rotation seeds. SO(3) denotes rotation-augmented training.

Training	Test frame	Sym. ↓	Angle ↓	IoU ↑
Canonical	Canonical	0.063	2.43°	0.864
Canonical	Rotated	0.595	53.10°	0.290
SO(3)	Canonical	0.090	4.46°	0.830
SO(3)	Rotated	0.195	13.55°	0.725

viding rotation equivariance. The primary-joint output does not model multiple joints, prismatic motion, or joint limits.

3.5. Discussion

The experiments indicate that semantic features and geometric aggregation play complementary roles in the current benchmark. Removing semantics worsens movable IoU, symmetric axis distance, and angular accuracy, supporting benefits for both part identification and hinge localization. The same-checkpoint global head already predicts a similar orientation, but its symmetric distance exceeds twice that of voting. This points to hinge-line localization as the main gain: direction alone does not determine hinge position. Canonical orientation remains a strong prior. Category-conditioned modes explain part of the angular accuracy, although the learned estimator improves upon them. Random rotations cause substantial degradation, and SO(3) augmentation only partly closes the gap. The supported scope is one primary revolute joint in canonically oriented semantic Gaussian assets derived from PartNet-Mobility; the evidence does not establish general articulation recovery for unconstrained real-world assets.

4. CONCLUSION

AxisField-GS recovers a primary revolute axis and movable field from a single-state semantic Gaussian asset using self-consistent weighted line estimation. Controlled readout comparisons and retraining ablations support geometric vote selection, particularly offset proximity. The results support feed-forward articulation of existing assets within the current limits of canonical orientation and a single revolute joint.

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